

Data Science & Artificial Intelligence

Machine Learning



Classification

DPP 01 Discussion Notes



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[NAT]

#Q. Find the mean of 0-1 loss for the given predictions:

Y	F(X)	loss
Cat	Cat	0
Cat	Dog	1
Dog	Panda	1
Panda	Panda	0
Rat	Dog	1
Rat	Rat	0

Round upto 1 decimal places

Sol. 0-1 loss fcn

if error = 1

if error = 0 loss = 0

So mean loss = $\frac{3}{6} = \frac{1}{2} = 0.5$

[MCQ]

#Q. What does the ROC curve represent?

- A** The trade-off between precision and recall.
- B** The relationship between accuracy and F1-score.
- C** The performance of a model across various thresholds.
- D** None of these

[MCQ]

#Q. Imagine you have a dataset that contains information about a binary classification model. The model classifies instances into two classes: "Positive" and "Negative". After evaluating the model, you have the following results:

True Positives (TP): 120

False Positives (FP): 25

True Negatives (TN): 800

False Negatives (FN): 30

Calculate the precision of the model using the given information.

$$\begin{aligned} \text{Precision} &= \text{TP} / \text{Total predicted P} \\ &= \text{TP} / (\text{TP} + \text{FP}) \\ &= 120 / 120 + 25 = 0.82 \end{aligned}$$

A 0.75

B 0.80

C ✓ 0.82

D 0.85

[NAT]

#Q. A machine learning model is trained to identify whether an email is spam or not. After evaluating the model on a test dataset, the following results are obtained

True Positive (TP): 175

True Negative (TN): 310

False Positive (FP): 123

False Negative (FN): 147

Calculate Precision?

$$\begin{aligned} \text{Sol. } P &= \frac{TP}{TP+FP} = \frac{175}{175+123} \\ &= 0.587 \end{aligned}$$

[MSQ]

#Q. How many statements are true?

* Type 1 error $\Rightarrow$ False Positive

Type 2 error $\Rightarrow$ False Negative A, C

- ☒ **A** Type I error: The number of instances incorrectly predicted as positive
- ☐ **B** Type II error: The number of instances correctly predicted as negative.
- ☒ **C** Type I error: The number of instances incorrectly predicted as negative.
- ☐ **D** Type II error: The number of instances incorrectly predicted as negative

[MCQ]

#Q. You are designing a deep learning system to detect driver fatigue in cars. It is crucial that your model detects fatigue, to prevent any accidents. Which of the following is the most appropriate evaluation metric?

Fatigue : Positive
Not Fatigue: Negative

• FP: driver is not Fatigue but Classifier say he is fatigue
≠ 0

• FN: driver fatigue, but Classifier say No
FN = 0

Recall = 1 } Recall = $\frac{TP}{TP + FN}$ we want FN = 0

A

Accuracy

B

Precision

☒

Recall

D

None of these

[MCQ]

#Q. Which of the following are supervised learning problems? (multiple may be correct)

↓
labelled data "y"

- ☒ **A** Learning to drive using a reward signal ← Supervised
- ☒ **B** Predicting disease from blood sample. ← supervised
- ☐ **C** Grouping students in the same class based on similar features. ← unsupervised
- ☒ **D** Face recognition to unlock your phone. ← Supervised

[MCQ]

#Q. Which of the following are classification problems? (multiple may be correct)

- ☒ **A** Predict the price of a house. → Regression
- ☒ **B** Predict the disease of a patient. → Category
- ☒ **C** Predict the temperature for tomorrow. → Regression
- ☒ **D** Has the student learned to play Guitar Yes/No
Learn to play a guitar. →

[MCQ]

#Q. What is the use of Validation dataset in Machine Learning?

↓
Ridge Regression

- A** To train the machine learning model.
- B** To evaluate the performance of the machine learning model
- C** ✓ To tune the hyperparameters of the machine learning model
- D** None of the above.

[MCQ]

#Q. Given a training data set of 10,000 instances, with each input instance having 17 dimensions and each output instance having 2 dimensions, the dimensions of the design matrix used in applying linear regression to this data is

- A** 10000×17
- B** 10002×17
- C** 10000×18 ✓
- D** 10000×19

X matrix
• design matrix

$$\begin{bmatrix} 1 & x \\ 1 & \\ \vdots & \\ 1 & \end{bmatrix}$$

10000×18

[MCQ]

#Q.

Logistic regression is robust to outliers. Why?

- ☒ **A** The squashing of output values between $[0, 1]$ dampens the affect of outliers.
- ☐ **B** Linear models are robust to outliers.
- ☐ **C** The parameters in logistic regression tend to take small values due to the nature of the problem setting and hence outliers get translated to the same range as other samples.
- ☐ **D** The given statement is false.

[MCQ]

#Q. Suppose I have 10,000 emails in my mailbox out of which 200 are spams. The spam detection system detects 150 mails as spams, out of which 50 are actually spams. What is the precision and recall of my spam detection system ?

- ☒ **A** Precision = 33.333%, Recall = 25%
- ☐ **B** Precision = 25%, Recall = 33.33%
- ☐ **C** Precision = 33.33%, Recall = 75%
- ☐ **D** Precision = 75%, Recall = 33.33%

Sol 200 Spam Actual
 150 Predicted Spam $\Rightarrow$ 50 True Spam
 100 False Spam

$$P = \frac{TP}{TP+FP} = \frac{50}{50+100} = \frac{1}{3}$$

$$R = \frac{TP}{\text{Total actual } P} = \frac{50}{200} = \frac{1}{4}$$

[MCQ]

#Q. Which of the following are true? TP True Positive, TN True Negative, FP False Positive, FN False Negative

A

$$\text{Precision} = \frac{TP}{TP+FP}$$

B

$$\text{Recall} = \frac{TP}{TP+FN}$$

C

$$\text{Accuracy} = \frac{2(TP + TN)}{TP+TN +FP+FN}$$

D

$$\text{Recall} = \frac{FP}{TP+FP}$$

[MCQ]

#Q. Suppose you are working on a binary classification problem using logistic regression. The logistic regression model is represented by the equation:

$$\text{Logit}(p) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where,

- $\text{Logit}(p)$ is the log-odds of the probability p belonging to the positive class.
- $x_1, x_2, \dots, x_n$ are the feature values.
- $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ are the coefficients.

Given the following logistic model:

$$\text{Logit}(p) = -2 + 1.5x_1 - 0.8x_2$$

Find out the probability p when $x_1 = 3$ and $x_2 = 2$.

Sol. $p = \frac{1}{1 + e^{-x_i \beta}} = \frac{1}{1 + e^{-(-2 + 1.5x_1 - 0.8x_2)}} = 0.711$

- A** 0.394
- B** 0.711 ✓
- C** 0.22
- D** 0.98

[MCQ]

#Q. The following data is to be used to construct a regression model:

X	10	10	10	10	10	10
Y	2	4	9	9	3	3

(data issue)

The value of the slope is_____.

- ☐ A .-2.167
- ☐ B -1.25
- ☐ C 1.25
- ☒ D None

Sol • $m = \frac{\text{Cov}(X, Y)}{\sigma_x^2}$

Here $\sigma_x^2 = 0$

[MCQ]

decision tree

#Q. Under what conditions does fitting a linear function in the leaf nodes of a regression tree instead of the constant function not increase the variance of the model?

Skip**A**

When the function is globally linear and each leaf node has at least 2 data points.

B

When the function is locally linear and each leaf node has at least 2 data points.

C

There is no change in model variance.

D

None of the above.

[MCQ]

#Q. In a classification dataset, there are 400 labeled examples belonging to 5 distinct classes. If each class has an equal number of examples, how many examples are there in each class?

• No of data points in a class = $\frac{400}{5}$

☒ **A** 80

☐ **C** 200

☐ **B** 100

☐ **D** 400

[NAT]

#Q. In a logistic regression model, ~~if the odds ratio for a predictor variable is 25,~~ what is the estimated increase in the odds of the outcome for a one-unit increase in the predictor variable?

$$\beta_1 = 2 \quad \beta_0 = 1$$

Sol $\log_e(\text{odds}) = (x_i \beta)$

$$\bullet \log_e(\text{odds}) = \beta_0 + \beta_1 x$$

$$\rightarrow \text{odds} = e^{\beta_0 + \beta_1 x}$$

$$\text{odds value} = e^{\beta_0 + \beta_1 x}$$

$$\text{if } x \text{ inc by } 1 \quad \text{odds value} = \underline{e^{\beta_0} + e^{\beta_1(x+1)}}$$

$$\frac{\text{odd new}}{\text{odd old}} = \frac{e^{\beta_0 + \beta_1(x+1)}}{e^{\beta_0 + \beta_1 x}} = e^{\beta_1}$$

$$\frac{\text{odd new}}{\text{odd old}} = e^{\beta_1}$$

$$\text{odd new} = e^2 \text{odd old}$$

✓

#Q. In a simple linear regression model, if the coefficient of determination (R^2) is 0.64, what percentage of the variation in the dependent variable is explained by the independent variable (in %)?

64%



THANK - YOU